matlab.mathworks.com

Verify it's you

HOME | PLOTS | APPS | FIGURE | FORMAT | TOOLS

Search (Ctrl+Shift+Space)

inbarasan

Show Code

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

FILE

LABELS AND ANNOTATIONS

/ › MATLAB Drive

Files

Name

- untitled5.m
- untitled4.m
- untitled3.m
- untitled2.m
- untitled01.m
- untitled.m mfw.m
- untitled.m

Workspace

Name	Value	Size	Class
d	1×50 double	1×50	double
f	[900,3500,28...	1×3	double
Freq	[900,3500,28...	1×3	double
Loss	[91.5249,103....	1×3	double
PL1	1×50 double	1×50	double
PL2	1×50 double	1×50	double
PL3	1×50 double	1×50	double

Editor

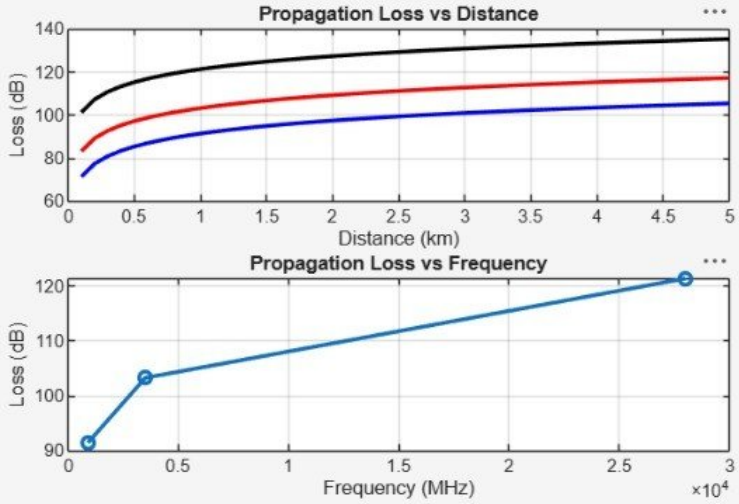
untitled.m | untitled2.m | untitled3.m | untitled4.m | untitled01.m | untitled5.m

/MATLAB Drive/untitled5.m

```
1   clc; clear; close all;
2   d = 0.1:0.1:5; % Distance (km)
3   f = [900 3500 28000]; % Frequency (MHz)
4   PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
5   PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
6   PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
7   figure
8   % -------- Graph 1 --------
9   subplot(2,1,1)
10
11  plot(d,PL1,'b','LineWidth',2); hold on
12  plot(d,PL2,'r','LineWidth',2)
13  plot(d,PL3,'k','LineWidth',2)
14  title('Propagation Loss vs Distance')
15  xlabel('Distance (km)')
16  ylabel('Loss (dB)')
17  grid on
18  % -------- Graph 2 (NO legend used to avoid error) --------
19  subplot(2,1,2)
```

Figures

Figure 1



Command Window

>>

matlab.mathworks.com

HOME | PLOTS | APPS | FIGURE | FORMAT | TOOLS

Search (Ctrl+Shift+Space)

inbarasan

Show Code

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

FILE

LABELS AND ANNOTATIONS

/ › MATLAB Drive

Files

Name

- untitled6.m
- untitled5.m
- untitled4.m
- untitled3.m
- untitled2.m
- untitled01.m
- untitled.m mfw.m
- untitled.m

Workspace

Name	Value	Size	Class
d	2	1×1	double
f	[900,3500,28...	1×3	double
PL	[97.5455,109....	1×3	double

Editor

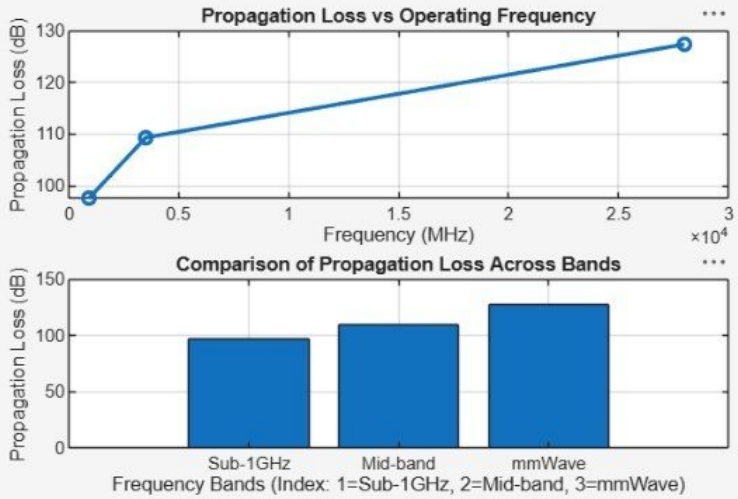
d3.m | untitled4.m | untitled01.m | untitled5.m | untitled2 | untitled6.m

/MATLAB Drive/untitled6.m

```
1   clc; clear; close all;
2   d = 2; % fixed distance (km)
3
4   f = [900 3500 28000]; % MHz
5   PL = 32.44 + 20*log10(f) + 20*log10(d);
6   figure
7   % -------- Graph 1: Frequency vs Loss --------
8   subplot(2,1,1)
9   plot(f,PL,'-o','LineWidth',2)
10  title('Propagation Loss vs Operating Frequency')
11  xlabel('Frequency (MHz)')
12  ylabel('Propagation Loss (dB)')
13  grid on
14  % -------- Graph 2: Band Comparison --------
15  subplot(2,1,2)
16  bar(PL)
17  title('Comparison of Propagation Loss Across Bands')
18  xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmW
19  ylabel('Propagation Loss (dB)')
```

Figures

Figure 1



Command Window

>>

HOME PLOTS APPS FIGURE FORMAT TOOLS

Search (Ctrl+Shift+Space)

inbarasan

ave As | Show Code

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

FILE

LABELS AND ANNOTATIONS

/ › MATLAB Drive

Files

Name

- untitled7.m
- untitled6.m
- untitled5.m
- untitled4.m
- untitled3.m
- untitled2.m
- untitled01.m
- untitled.m mfw.m
- untitled.m

Workspace

Name	Value	Size	Class
d	2	1×1	double
f	[900,3500,28...	1×3	double
PL	[97.5455,109....	1×3	double

Editor

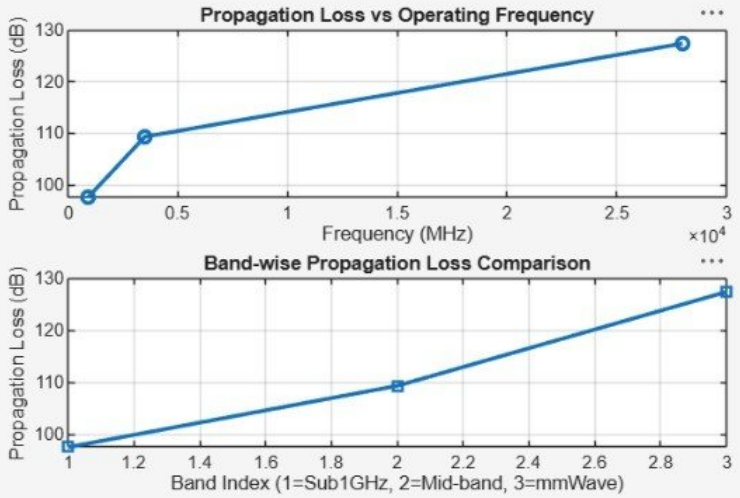
d4.m | untitled01.m | untitled5.m | untitled2 | untitled6.m | untitled7.m

/MATLAB Drive/untitled7.m

```
clc; clear; close all;
% Frequency bands (MHz)
f = [900 3500 28000];
% Distance (km)
d = 2;
% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Simple Line Plot (SAFE) --------
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Safe Comparison Plot --------
subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
```

Figures

Figure 1



Command Window

>>